



SYAHRUL ARRIZAL MA'ARIF

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Banyumas, Central Java

Bachelor's degree graduate in Electrical Engineering with a specialization in Instrumentation and Control, equipped with hands-on experience in automation within the manufacturing and technology sectors. Currently working as **Jr. Manufacturing Engineer in Fast Moving Consumer Good (FMCG)** company. Strong technical knowledge & experience as basic manufacturing sectors increase production manufacturing performance as a team or individual.

EDUCATION

Bachelor Degree in Electrical Engineering, Jenderal Soedirman University 2020 – 2024

- Current GPA: 3.80/4.00 (158 credits)

EXPERIENCES

-
- Jr. Manufacturing Engineer, PT Smoore Technology Indonesia** Nov 2024 - Present
- Monitoring and control production achievement related to OEE (Overall Equipment Effectiveness), Output, Yield (Quality), and overall efficiency.
 - Lead, monitoring, & improvement activity to increase performance & minimize downtime or waste.
 - Involved in production process improvements using Lean Six Sigma (DMAIC), 7 QC Tools, Root Cause Analysis, and 5S to enhance quality and reduce waste.
 - Conducted data analysis and reporting to support output, quality control, defect tracking, and process optimization.
 - Collaborated with production and quality teams to identify and resolve equipment-related issues through root cause analysis and implementation of CAPA.
 - Collaborated with production to increase production efficiency, UPH, & UPPH.
- Internship Hardware Engineer, MBKM PT Stechoq Robotika Indonesia.** 2023
- Designing PLC programs algorithm and SCADA system.
 - Research and development sheet metal bending automation
 - Manufacturing and assembly
 - As project manager, has a responsibility to leading, controlling, and evaluating team project.
- Laboratory Assistant of Industrial Otomation at Electrical Engineering Unsoed** 2023
- Teach practical material with Programming Logic Controller (PLC)
 - Assess and correct participant performance
 - Assess and correct participants' final project
- Laboratory Assistant of Instrumentation at Electrical Engineering Unsoed** 2023
- Teach practical material with Microcontroller and Instrumentation Components
 - Assess and correct participant performance
 - Assess and correct participants' final project
- Mechanical Engineer, UKM Soedirman Robotic Team** 2022
- Designing 3D drawing lagori pile robot
 - Designing algorithm and system
 - Manufacturing and assembly
- Staff of Kementerian Kaderisasi dan Pengembangan Mahasiswa, BEM FT** 2022
- Responsible for conducting student development of the Faculty of Engineering
 - Preparing character regeneration of new student Faculty of Engineering
 - As secretary of division, responsible to legality and administration of division.
- Head of Pengembangan dan Sumber Daya Manusia, UKM Senior Faculty of Engineering** 2022
- Preparing new generation of member UKM Seni dan Olahraga (Senior)
 - Controlling and evaluating all member and division
 - Responsible to member skill development

Secretary Committee , Pengenalan Kehidupan Kampus Mahasiswa Baru (PKKMB) Faculty of Engineering	2022
• Responsible to legality and administration of event • Responsible to participant attendance	
Head of Field Coordinator Commitee, School Of Leadership Faculty of Engineering	2022
• Prepare all the field equipment of event • Arrange work field system of event • Controlling and evaluating to ensure the event run on schedule and plan	
Staff of Human Resource Departement, UKM Sodirman Robotic Team	2021
• Responsible to human skill development • Controlling and evaluating all member to ensure work in their responsibilities	
Staff of Pendamping Kelompok Committee, Pengenalan Kehidupan Kampus Mahasiswa Baru (PKKMB) Faculty of Engineering	2021
• Guiding and Controlling the participant • Giving basic knowledge about faculty of engineering	
Staff of Kaderisasi Committee, School Of Leadership Faculty of Engineering	2021
• Responsible to spread basic knowledge about leadership • Controlling and ensure the participant to practice in project team	

PROJECT

Work Standardization to reduce downtime due to MSA frequency

- This project aims to reduce downtime caused by frequent Measurement System Analysis (MSA) requirements in the product detection system, which is currently impacted by high product variation. To address this, we will develop a standardized approach using the DMAIC (Define, Measure, Analyze, Improve, Control) methodology. The solution will consider machine capability and ensure alignment with Critical to Quality (CTQ) requirements.

Headcount Reduction and Output Improvement by Automating the Detection System

- This project aims to reduce headcount and improve operational efficiency by upgrading the current manual detection system to an automated detection system. The automation will streamline processes, minimize human intervention, and enhance overall productivity.

Sheet Metal Bending Automation

- In this project, we are development system of sheet metal bending. In this case the sheet metal bending machine still uses a manual system in its operation. We were assigned to make the automation system. We designed this system with Omron PLC. In the automation, it uses an algorithm designed on a ladder diagram in the CX programmer software.

Lagori Pile Robots

- This project is designed for KRI 2022 activities. In this year's theme, robots are required to build a system that can knock down piles and be able to rearrange them. The mechanical design of the ejection robot uses a double motor and the robot that composes it uses a pulley. This needs to be designed in 3D design in the Solidwork 2020 software

Smart Alarm Fire Warning

- Smart alarm fire warning is simple internet of things project. This project using flame sensor, gas sensor, and also using buzzer actuator. when the sensor detects a fire, the signal sent by the nodemcu 8266 microcontroller will be transferred remotely to telegram via the server and the buzzer will sound as a warning.

Smart IoT Aquaculture (Water Quality and Controlling of Fish Tank)

- Water monitoring and controlling system in ornamental fish aquariums with pH and temperature parameters, and equipped with cooling actuators, heaters, and a pH balancing system.

LANGUAGES

- Indonesia
- English

SKILLS

Hardskill

- Electrical Automation Control System
- PLC, SCADA, & HMI Programming
- Corrective & Preventive Maintenance
- Wiring Diagrams & Technical Drawings
- Work Standardization
- Overall Equipment Effectiveness (OEE)
- Root Cause Analysis (RCA)
- Lean Six Sigma & 7QC Tool
- Failure Mode & Effect Analysis (FMEA)
- Microsoft Family Software

Softskill

- Leadership
- Problem Solving
- Effective Communication
- Adaptability
- Team Work

Certificate

- White Belt Lean Six Sigma Course Certification
- Yellow Belt Lean Six Sigma Course Certification
- Python Course at Progate
- Python Fundamental for Data Science at DQ Lab
- Introduction to Data Science With Python at DQ Lab
- Data Wrangling at DQ Lab
- Basic python at Skilvul (Samsung Innovation Campuss Program)
- Intermediate Python at Skilvul (Samsung Innovation Campuss Program)

Award

- 2nd Winner at business idea competition by Faculty Of Animal Husbandry, Brawijaya University with theme “National Innovation in Livestock Technology and Software” 2023
- Best Project Manager by MBKM Internship at PT Stechoq Robotika Indonesia 2023